CLAIMS

What is claimed is:

1. A method for high-bandwidth data transfer between two uncoupled mobile electronic devices, the method comprising the steps of:

    - displaying, on a screen of a first mobile device, a sequence of data frames, wherein each frame encodes a portion of a data set into a two-dimensional grid of channel elements, and wherein each channel element comprises a plurality of physical pixels all set to a substantially uniform emission characteristic;
    - capturing, using an image sensor of a second mobile device, said sequence of data frames over a free-space optical path; and
    - decoding, by a processor on the second mobile device, the captured sequence of frames to reconstruct the data set.

2. The method of claim 1, wherein the emission characteristic is color.

3. The method of claim 1, further comprising an initial step of displaying a synchronization pattern on the screen of the first device and capturing said pattern with the image sensor of the second device to establish data transfer parameters.

4. The method of claim 3, wherein the synchronization pattern is a QR code containing at least one of: the dimensions of the channel element grid, the data rate, or an error correction scheme.

5. The method of claim 1, wherein the size of the channel elements is dynamically adjusted based on the distance or alignment between the two devices.

6. A system for high-bandwidth data transfer, comprising:

    - a first mobile device configured to:
        - encode a data set into a sequence of frames, wherein each frame is a grid of logical channel elements that are larger than the physical pixels of a display screen; and
        - display said sequence of frames on the display screen;
    - a second mobile device configured to:
        - capture the sequence of frames using its camera; and
        - decode the sequence of frames to reconstruct the data set.

7. The system of claim 6, wherein the first mobile device is further configured to initiate the transfer by displaying a machine-readable synchronization pattern.

8. The system of claim 6, wherein the data transfer rate is at least 1 gigabit per second.

9. A non-transitory computer-readable medium containing instructions that, when executed by a processor on a mobile device, cause the device to perform the steps of:

    - receiving a data set to be transmitted;
    - generating a sequence of visual data frames, wherein each frame represents a portion of the data set encoded as a grid of colored blocks, each block comprising a plurality of physical screen pixels; and
    - causing a display screen of the mobile device to render the sequence of visual data frames in rapid succession.

10. The medium of claim 9, wherein the instructions further cause the device to first display a synchronization pattern containing metadata required for a receiving device to decode the subsequent data frames.